Motor Torque Fault Diagnosis for Four Wheel Independent Motor-Drive Vehicle Based on Unscented Kalman Filter

Hongliang Zhou, Zhiyuan Liu, and Xingwang Yang

***Abstract*—An motor torque fault diagnosis for four wheel independent motor-drive vehicle is presented in this paper. Focusing on the fault that the actual motor torques differ from the motor torques commanded by vehicle control unit, the fault parameters are introduced to represent thedifferences and a nonlinear vehicle dynamics model is built. This model considers vehicle body nonlinear dynamics, wheel dynamics, and tire nonlinear characteristics, and the fault parameters are set to be the coefficients of motor torque commands. Then the motor fault diagnosis problem is converted to these fault parameters identification problem, and unscented Kalman filter is implemented to identify them in real time. With the road test data of a four wheel independent motor-drive vehicle, the fault diagnosis method is testified to be effective both in normal and extreme handling situations.**

***Index Terms*—Nonlinear vehicle dynamics, fault diagnosis, unscented Kalman filter, four wheel independent motor-drive vehicle.**

NOTATIONS

v_x	Vehicle longitudinal speed in body-fixed coordinate system.
v_y	Vehicle lateral speed in body-fixed coordinate system.
v	Vehicle speed.
r	Vehicle yaw rate.
β	Vehicle side slip angle.
δ	Steering angle.
m	Vehicle mass.
J_z	Vehicle yaw moment of inertia.
l_f	Distance from vehicle center of gravity to front axis.
l_r	Distance from vehicle center of gravity to rear axis.
d	Track width of vehicle.
F_{xi}	Tire/road longitudinal force of each wheel.
F_{yi}	Tire/road lateral force of each wheel.
F_{ti}	Tire/road force of each wheel.
λ_i	Wheel slip of each wheel.
S_{Resi}	Resultant wheel slip of each wheel.
α_i	Wheel slip angle.
ω_i	Rotational speed of each wheel.
k	Tire stiffness coefficient.
v_{xi}	Wheel ground contact point velocity.
T_i	Motor torque.
f_i	Fault parameter of each motor.
J_ω	Wheel ratation moment of inertia.
R	Wheel radii.
i	1, 2, 3, 4 as the index of four wheels.

Abbreviation

UKF	Unscented Kalman Filter.
4WIMD	Four wheel independent motor-drive.
CoG	Center of gravity.
DoF	Degree of Freedom.
VCU	Vehicle control unit.
MCU	Motor control unit.
PMSM	Permanent magnet synchronous motor.
CAN	Controller area network.
E/E	Electric and electronic architecture.

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I. INTRODUCTION

WITH the benefits of driveability and energy saving, the four wheel independent motor-drive (4WIMD) electric vehicle is a promising vehicle driveline structure, which includes in-wheel motor-drive [1] and wheel-side motor-drive [2]. As the torques on each wheel can be controlled independently, the vehicle dynamics stability can be enhanced by allocating the motor driving or re-generative torques on each wheel[3-5]. So the key factor of vehicle performance is to control motor output torque accurately. In a conventional 4WIMD vehicle, the electric and electronic (E/E) architecture is usually designed in the way as:

1) The vehicle control unit (VCU) is the core controller of the whole E/E system to acquire signals of sensors, such as accelerator pedal and brake pedal position, yaw rate, accelerations, etc., to estimate vehicle states, to make the decision on the torque values of each motor and then to send the torque commands to motor control units (MCUs) via controller area network (CAN).

2) The MCUs are in charge of controlling motor output torques to meet the above torque commands, and upload the motors status, such as actual torques, rotation speeds, current, temperature, fault status, etc.

The MCUs monitor the motor status according to motor voltage, current, speed, temperature, etc., it is complicated and it should be reliable in most situations [6]. However, as the driveline is so important and relates to vehicle safety, it is also required there are redundant diagnosis routines in VCU. Reference [7] analyzes the sources of faults in vehicle subsystem and classifies them to different groups. For the driveline with in-wheel motors, the faults include three-phase balanced short circuit, inverter shutdown, current sensor misalignment and so on. Through simulation, the vehicle status is analyzed under faults condition, and the results show that there are undesired yaw rate, lateral acceleration and longitudinal acceleration and the vehicle is hard to control. Reference [8] develops a combinational-logic fault diagnosis algorithm with simple combination logic and threshold to catch motor fault. Reference [9] presents an active fault diagnosis for 4WIMD vehicle. The motor fault is introduced as control gain, and it is defined according to vehicle driving situation, which is straight line case and turning case. And the method is testified with CarSim simulation software. Based on reference [9], the team presents an improved motor torque fault modeling method including torque loss and additive torque, and disturbance observer to observe actual torque and recursive least-squares estimator to determine the torque loss and additive torque [10]. Reference [11] also introduces a gain to model the motor fault, and sliding model control method is used as fault-tolerant control. But the gain obtaining method is not included.

In this paper, a motor torque fault diagnosis method is presented based on Unscented Kalman Filter (UKF). Firstly, a 7 degree-of-freedom (DoF) nonlinear vehicle dynamics model is built, which includes vehicle body nonlinear dynamics, wheel dynamics and tire nonlinear characteristics. In the model, the fault parameters are introduced, which are the coefficients between the actual motor torques and torque commands and are unknown. So the fault diagnosis problem is converted to fault parameters identification problem. Based on the information from vehicle sensor and CAN bus, the UKF is implemented to estimate vehicle states and the above fault parameters in real-time. The states of nonlinear vehicle model are augmented with these fault parameters, which is modeled in random walking way. This proposed fault diagnosis algorithm is tested in experiments with a 4WIMD vehicle. In normal vehicle driving situation, the fault parameters are estimated in high precision. In extreme vehicle driving situation, as the motor torques change severely, the estimated fault parameters oscillate two much and need to be filtered. The tests results show that the proposed method is effective and implementable.

The rest of this paper is organized as following. The 4WIMD vehicle and motor torque fault diagnosis problem are introduced in Section II. The vehicle nonlinear model with fault parameters is built in Section III-A. The fault diagnosis algorithm based on UKF is designed in Section IV. Evaluation results are illustrated in Section V, followed by the conclusion in Section VI.

Fig. 1. Four wheel indepedent motor-drive vehicle.

II. 4WIMD Vehicle and Motor Fault Diagnosis Problem

The vehicle considered in the paper is a 4WIMD vehicle, which is designed by First Automotive Works Group Corporation of China and is shown in Fig 1. The vehicle is equipped with four permanent magnet synchronous motors (PMSMs) besides the four wheels. For the E/E architecture of the driveline, there is a powerful VCU and four MCUs, and they are connected by CAN bus. The motor torques are commanded by VCU and the motor torques, speed, status are reported from MCUs. According to driver manipulation and vehicle states, the motor torques can be positive (driving) and negative (re-generative). Furthermore, an inertial sensor cluster is equipped at the vehicle center of gravity (CoG) to test vehicle longitudinal acceleration, lateral acceleration and yaw rate.

In normal condition, the actual motor torques are approaching the VCU's torque commands with small delay and minor errors. MCUs calculate the motor torques based on the output of the current sensor and PMSMs parameters. However, if there are faults in PMSMs and MCUs, the actual torques would deviate the torque commands and the torques reported by the MCUs maybe also have basis. In this condition, if VCU control the vehicle without the knowledge of the basis, it will reduce the vehicle driveability and stability, and even cause serious accidents. For the reason of safety, it is required to develop a fault diagnosis algorithm for VCU to monitor motor torques.

III. Vehicle Nonlinear Model With Motor Faults

A. Vehicle Body Nonlinear Model

The vehicle body dynamics is complicated with longitudinal velocity, lateral velocity, yaw rate, roll rate, pitch rate and the coupling of each other. Besides the tire/road forces, the aerodynamics drag force is another input. To simplify the fault diagnosis algorithm, a 3 DoF nonlinear vehicle body dynamics model is built in this section. By ignoring the roll, pitch dynamics and aerodynamic drag force, the vehicle longitudinal velocity v_x, lateral velocity v_y and yaw rate r in vehicle body-fixed coordinate system are considered as model states, shown in Fig. 2 and (1). F_{xi}, F_{yi}, α_i are tire longitudinal, lateral forces and wheel slip angle. δ is front wheel steering angle and β is vehicle body

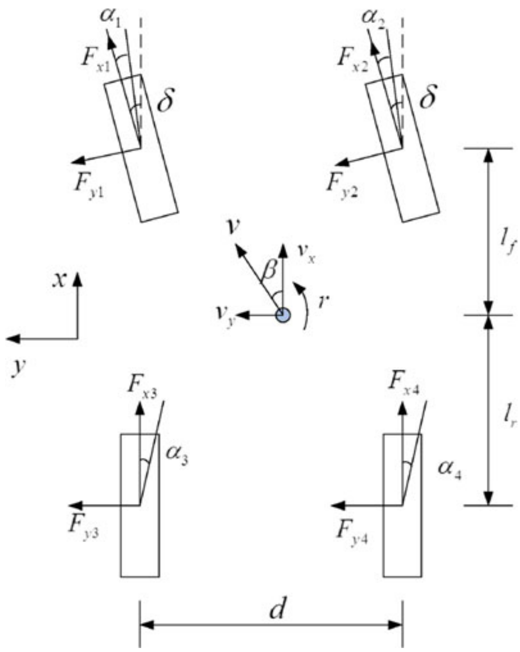

Fig. 2. Vehicle body nonlinear dynamics.

slip angle. l_f, l_r and d are the distance from front wheel axle, rear axle to vehicle center of gravity and wheel distance. m is vehicle mass and J_z is vehicle yawinertia.

$$
\begin{aligned}
\dot{v}_x &= r v_y + \frac{1}{m}\left((F_{x1}+F_{x2})\cos\delta - (F_{y1}+F_{y2})\sin\delta \right.\\
&\quad \left. + F_{x3} + F_{x4}\right)\\
\dot{v}_y &= -r v_x + \frac{1}{m}\left((F_{y1}+F_{y2})\cos\delta + (F_{x1}+F_{x2})\sin\delta \right.\\
&\quad \left. + F_{y3} + F_{y4}\right)\\
\dot{r} &= \frac{1}{J_z}\Bigg(l_f(F_{x1}+F_{x2})\sin\delta + l_f(F_{y1}+F_{y2})\cos\delta\\
&\quad - l_r(F_{y3}+F_{y4}) + \frac{d}{2}(F_{x2}-F_{x1})\cos\delta\\
&\quad + \frac{d}{2}(F_{x4}-F_{x3}) + \frac{d}{2}(F_{y1}-F_{y2})\sin\delta\Bigg)
\end{aligned} \tag{1}
$$

The wheel slip angles are calculated as (2).

$$
\begin{aligned}
\alpha_1 &= \delta - \frac{v_y + l_f r}{v_x - \frac{d}{2} r}\\
\alpha_2 &= \delta - \frac{v_y + l_f r}{v_x + \frac{d}{2} r}\\
\alpha_3 &= -\frac{v_y - l_f r}{v_x - \frac{d}{2} r}\\
\alpha_4 &= -\frac{v_y - l_f r}{v_x + \frac{d}{2} r}
\end{aligned} \tag{2}
$$

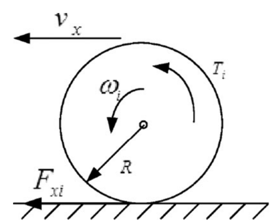

Fig. 3. Wheel dynamics.

B. Wheel Dynamics With Motor Fault

Considering the wheel rolling resistance is small comparing to motor torques, it is ignored in wheel dynamics modeling. The wheel dynamics model is shown in Fig. 3 and (3), where ω_i $(i = 1, 2, 3, 4)$ are wheel rotation velocities, R is tire rotation radii, J_w is wheel rotation inertia, and T_i are motor output torques ($T_i > 0$ for motor driving torques and $T_i < 0$ for motor regenerative braking torques).

$$
\dot{\omega}_i = \frac{1}{J_w}(-F_{xi}R + T_i) \tag{3}
$$

Just as the introduction in Section I, the VCU is the core to decide torques of each wheel and to send the torque commands. The motor and its controller are considered as one single module with torque command as input and actual torque as output. Due to sensor fault, open circuit fault, short circuit fault or even parameters variation, the actual motor torque is not the same as the torque command. If the difference is too large, the controllability of vehicle decrease. That is the motor fault considered in this paper, so the fault parameters f_i are introduced and are defined as (4) [9], [10]. These fault parameters are continuous and some situations are listed below:

1) When the motors are normal, the motor torques T_i are equal to torque command T_{ci} and f_i are 1;

2) When the motors are total failure, T_i are 0 and f_i are 0;

3) When actual motor torque T_i is less than torque command T_{ci}, f_i is less than 1, and f_i is larger than 1 when T_i is larger than T_{ci}.

$$
T_i = f_i T_{ci} \tag{4}
$$

In (3), F_{xi} are decided by wheel slip λ_i. According to Reference [12], λ_i are effected by vehicle yaw rate r and wheel slip angle α_i and are calculated as (5). When $\omega_i R\cos\alpha_i \geq v_{xi}$, the wheels are in driving mode and $T_i \geq 0$. Otherwise, the wheels are in braking mode and $T_i < 0$

$$
\lambda_i = \begin{cases} \dfrac{\omega_i R\cos\alpha_i - v_{xi}}{\omega_i R\cos\alpha_i} & \omega_i R\cos\alpha_i \geq v_{xi}\\[2ex] \dfrac{\omega_i R\cos\alpha_i - v_{xi}}{v_{xi}} & \omega_i R\cos\alpha_i < v_{xi} \end{cases} \tag{5}
$$

In (5), v_{xi} are the wheel ground contact point velocities, and are calculated as:

$$\begin{aligned} v_1 &= \sqrt{{v_x}^2 + {v_y}^2} - r\left(\frac{d}{2} - l_f\frac{v_y}{v_x}\right) \\ v_2 &= \sqrt{{v_x}^2 + {v_y}^2} + r\left(\frac{d}{2} + l_f\frac{v_y}{v_x}\right) \\ v_3 &= \sqrt{{v_x}^2 + {v_y}^2} - r\left(\frac{d}{2} + l_r\frac{v_y}{v_x}\right) \\ v_4 &= \sqrt{{v_x}^2 + {v_y}^2} + r\left(\frac{d}{2} - l_r\frac{v_y}{v_x}\right) \end{aligned} \tag{6}$$

C. Tire Model

In the vehicle body nonlinear model (1) and wheel dynamics model (3), tire forces F_{xi} and F_{yi} are inputs and are decided by wheel slip λ_i, wheel slip angle α_i, road adhesion coefficient μ and wheel load F_{zi}. The relationship is complicated and some nonlinear models are designed to describe the nonlinear characteristics, In the paper, a linear tire model is chosen and the model is as:

$$F_{ti} = k s_{Resi} \tag{7}$$

In (7), F_{ti} is the resultant of tire longitudinal and lateral forces, s_{Res} is the resultant of wheel longitudinal slip and slip angle, and k is the tire stiffness coefficient. s_{Resi} is calculated as (8) and the longitudinal and lateral forces are calculated as (9).

$$s_{Resi} = \sqrt{{\lambda_i}^2 + {\alpha_i}^2} \tag{8}$$

$$\begin{aligned} F_{xi} &= \frac{\lambda_i}{s_{Resi}} F_{ti} \\ F_{yi} &= \frac{\alpha_i}{s_{Resi}} F_{ti} \end{aligned} \tag{9}$$

D. Vehicle Nonlinear Model

According to Section III-A to III-C, the vehicle nonlinear model is built as:

$$\dot{x} = f(x) + Bu \tag{10}$$

where $x = [v_x,\ v_y,\ r,\ \omega_1,\ \omega_2,\ \omega_3,\ \omega_4]^T$ are model states, $u = [T_{c1},\ T_{c2},\ T_{c3},\ T_{c4}]^T$ are control inputs, and B is the control input matrix constructed from (1), (3) and (4). $f(x)$ is nonlinear function, which can be easily obtained according to (1) to (9) and is not described for simplicity in the paper. In this model, δ is considered to be time-varying parameters.

According to the sensors and information described in Section II, the outputs of the vehicle nonlinear model are $y = [a_x,\ a_y,\ r,\ \omega_1,\ \omega_2,\ \omega_3,\ \omega_4]^T$. a_x, a_y and r are vehicle longitudinal acceleration, lateral acceleration and yaw rate, which are the outputs of inertial sensor cluster. ω_i are reported to VCU by MCUs. And their relationship to vehicle states x are shown in (11).

$$y = h(x) \tag{11}$$

where

$$h(x) = \begin{bmatrix} \frac{1}{m}\left((F_{x1} + F_{x2})\cos\delta - (F_{y1} + F_{y2})\sin\delta \right. \\ \left. \quad + F_{x3} + F_{x4}\right) \\ \frac{1}{m}\left((F_{y1} + F_{y2})\cos\delta + (F_{x1} + F_{x2})\sin\delta \right. \\ \left. \quad + F_{y3} + F_{y4}\right) \\ r \\ \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix}$$

IV. Fault Parameters Identification Based on UKF

According to the fault diagnosis problem in Section II and the vehicle model in Section III, the parameters f_i should be estimated in real time for motor fault diagnosis. The estimation method chosen in the paper is UKF method. UKF is a nonlinear filter, which is presented by Julier etc. in 1995 [13]. it is a kind of recursive Bayesian filer and it is based on a statistical linearization technique. Starting with a nonlinear function of random variables, a linear regression between n points is drawn from the prior distribution of the random variables. This technique has a higher order of accuracy and gives a more accurate result than analytical linearization techniques, such as Taylor series linearization, as it considers the spread of the random variables. For linear system, the performance of UKF is same as the performance of extended Kalman filter (EKF), but for nonlinear system, the performance is much better than that of EKF. As there is no need to linearize the nonlinear system function and output function of system model by calculating the Jacobian and Hessians matrices, it is free of truncation error of higher order term [14], [15]. In [16], EKF, UKF, adaptive EKF and adaptive UKF used in on-line estimation of model parameters and state-of-charge (SoC) of Lithium-Iron batteries are compared and the results show that the performance of adaptive UKF is better than adaptive EKF and the performance of UKF is better than EKF, as the statistics of voltage errors are 0.06 V, 0.09 V, 0.11V and 0.16 V and the statistics of SoC are 0.10%, 0.22%, 0.47%, and 0.71% respectively. Some other comparison of UKF and EKF in different implementations are shown in [17]–[19].

In order to identify the fault parameters f_i with UKF, the states of vehicle model is extended as $\tilde{x} = [v_x, v_y, r, \omega_1, \omega_2, \omega_3, \omega_4, f_1, f_2, f_3, f_4]^T$ and the dynamics of f_i is described by random walking model. Based on (10) and (11), to apply UKF, the vehicle model is discretized with Euler method and is re-constructed as (12).

$$\begin{cases} \tilde{x}(k+1) = \tilde{f}(\tilde{x}(k)) + T_s \widetilde{B} u(k) + v(k) \\ y(k) = \tilde{h}(\tilde{x}(k)) + w(k) \end{cases} \tag{12}$$

$v(k)$ and $w(k)$ are the process noise and measurement noise respectively, and they are considered to be non-intercorrelated, stationary, white and Gaussian with known covariance of Q_v and R_w. $\widetilde{f}(x(k))$ is a column vector with 11 items and is constructed with two kinds of items. The first 7 items are $\tilde{f}(x(k))_{[1:7]} = x(k) + T_s f(x(k))$ and T_s is the sampling period. The last

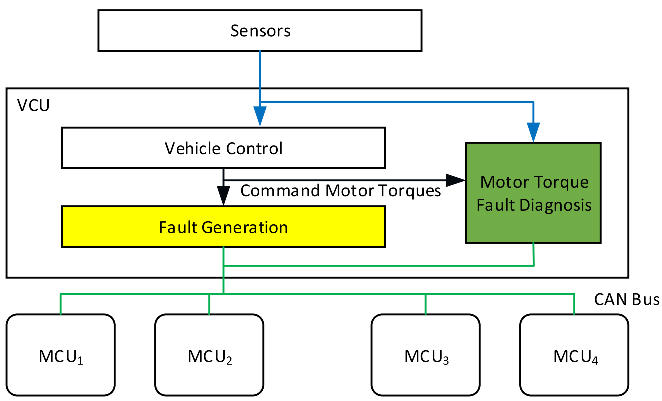

Fig. 4. Motor torque fault diagnosis algorithm evaluation structure.

4 items are $\tilde{f}(x(k))_{[8:11]} = [f_1(k), f_2(k), f_3(k), f_4(k)]^T$. The input matrix is

$$\widetilde{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ f_1 & 0 & 0 & 0 \\ 0 & f_2 & 0 & 0 \\ 0 & 0 & f_3 & 0 \\ 0 & 0 & 0 & f_4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{13}$$

The structure of the output function $\tilde{h}(x(k))$ is as $\tilde{f}(x(k))$, with the first 7 items as $\tilde{h}(x(k))_{[1:7]} = h((x(k)))$ and the last 4 items as $\tilde{h}(x(k))_{[8:11]} = [0, 0, 0, 0]^T$.

The UKF recursion process is omitted in the paper, please refer to [20].

V. Fault Diagnosis Evaluation Based on Vehicle Road Tests

The evaluation structure is shown in Fig. 4. In VCU, there are three main parts. The "Vehicle Control" part is the core control algorithm and it calculates the command motor torques according to sensors, which are acceleration pedal position sensor, brake pedal position sensor, inertia sensor cluster and so on. The "Fault Generation" part is to simulate motor torque faults as it is hard to test diagnosis algorithm with real motor faults. It receives the motor torques commands, multiple them with designed fault parameters and then send the faulty torques to MCUs through CAN bus, which makes it easy to simulate motor faults without danger. The "Motor Torque Fault Diagnosis" part is the algorithm proposed in this paper and it gets vehicle body dynamics through sensor signals and wheel dynamics through CAN bus, and estimates motor torque fault parameters. It should be noticed that there is no fault tolerant control algorithm in VCU, as this algorithm usually sets the faulty motor torque to 0 to prevent unstable vehicle states, which makes it impossible for motor torque fault diagnosis.

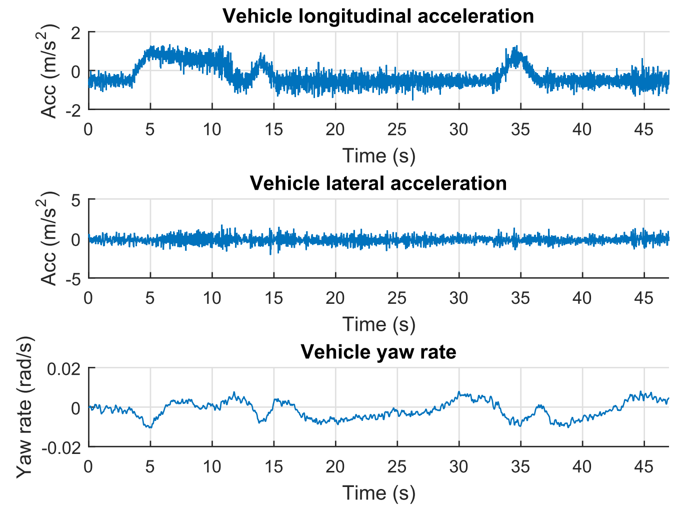

Fig. 5. Vehicle inertia sensor cluster outputs in straight driving situation.

To evaluate the proposed motor torque fault diagnosis algorithm, two tests with the 4WIMD electric vehicle shown in Fig. 1 are carried out according to typical vehicle driving situations. To cover vehicle driving status and fully evaluate the diagnosis algorithm, the first test is designed as normal straight driving situation and the second test is designed as critical cornering situation. In these tests, the initial value of fault parameters $\hat{f}_i$ are set to 1 to indicate normal motor state. The initial value of $P \in R^{11\times 11}$ is set to $P = diag(0.00005, 0.00005, 0.00005, 0.00005, 0.00005, 0.00005, 0.00005, 0.00005, 0.00005, 0.00005, 0.00005)$. The system noise $Q_v \in R^{11\times 11}$ is set to $Q_v = diag(0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001)$, and the measurement noise $R_w \in R^{7\times 7}$ is set to $R_w = diag(0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001, 0.00001)$. The sampling period is 0.005 s.

A. Vehicle Normal Straight Driving Situation

In this situation, the vehicle runs on a high adhesion coefficient road and the "Fault Generation" part sets the motor torque on front right wheel to be 0 to simulate this motor failure. During this test, the vehicle is running to right as there is no torque acting on front right wheel. It is the responsibility of the driver to steer the vehicle to follow the lane in the test field.

The vehicle body inertia sensor cluster outputs and wheel speeds are shown in Figs. 5 and 6. The vehicle runs in straight line as lateral acceleration and yaw rate are both very small. In the time ranges of 3.6 s–11.5 s and 33.4 s–36.0 s, vehicle is accelerating as the longitudinal acceleration is positive. As the actual torques in the front right wheel is set to 0, the wheel speed of this wheel is a little smaller than the other three wheel speeds in accelerating situation, which is the solid black curve and can be seen in the enlarged figure in Fig. 6.

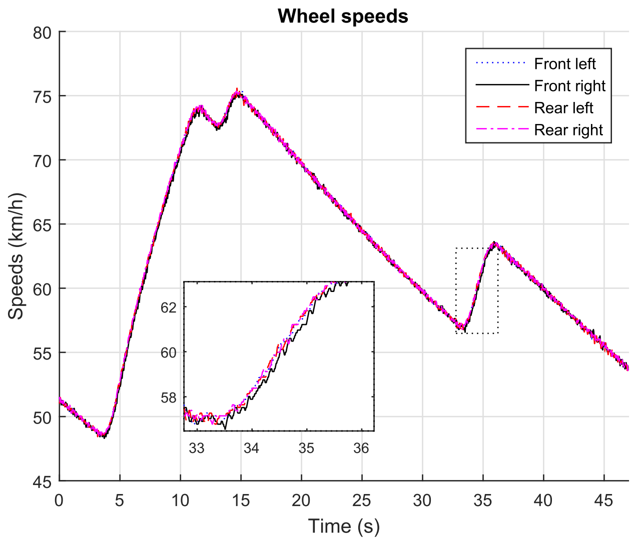

Fig. 6. Wheel speeds in straight driving situation.

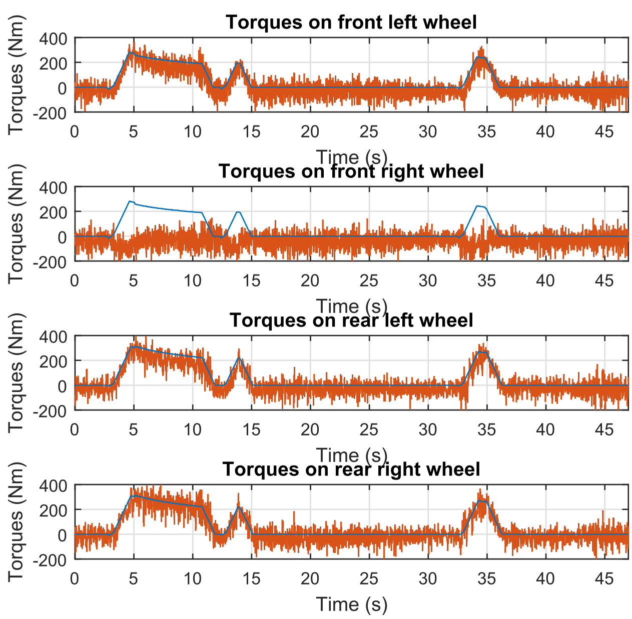

Fig. 7. Motor torque and road/tire torques in straight driving situation. (The blue lines are commanded motor torques and red lines are the products of tire/road forces and wheel radii).

According to Section IV, the vehicle longitudinal and lateral speeds are estimated based on UKF, so the tire/road longitudinal forces of each wheel can be calculated according to (2), (5), (7), (8) and (9). By multiplying wheel radii, the tire/road longitudinal forces and commanded motor torques are shown in Fig. 7. The tire/road force of front right wheel is almost 0 in the whole test process. For the other three wheels, the products of tire/road forces and radii are a bit smaller than motor torques because of the ignorance of wheel rolling resistance.

The fault parameters of each wheel estimated by the fault diagnosis algorithm are shown in Fig. 8. The labels A, B and C indicate three different situations during the test. Referring to above figures, these situations are explained as:

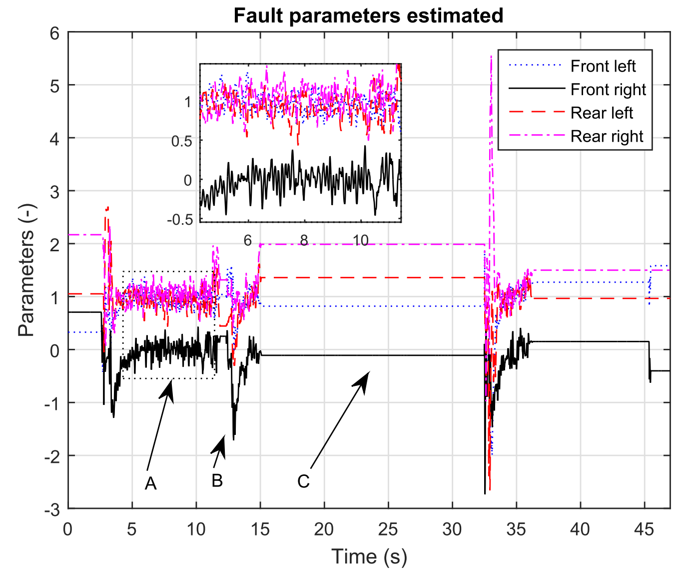

Fig. 8. Fault parameters estimated in straight driving situation.

1) Label A indicates the situation that vehicle accelerates and the motor torques are positive. The fault parameters $\hat{f}_i$ of front left, rear left and rear right wheels are around 1 and fault parameters $\hat{f}_2$ of front right wheel is around 0. The estimation of fault parameters is accurate for both the normal and fault motors.
2) Label B indicates the situation that motor torques change rapidly, including motor torques increasing sharply (referring to the time range of 3 s to 4 s) and motor torques fluctuating (referring to the time range of 11s to 15 s and 32 s to 36 s). The fault parameters $\hat{f}_i$ fluctuating sharply too. However, $\hat{f}_i$ of front left, rear left and rear right wheels are in the same range and $\hat{f}_2$ is smaller than the other three $\hat{f}_i$.
3) Label C indicates the situation that motor torque commands T_{ci} are 0. The value of fault parameters f_i are not updated and kept the values when the torques became 0. In this situation, the state of motor could not be diagnosed.

In the vehicle straight normal driving situation, the motor torque fault parameters f_i can be estimated by the proposed fault diagnosis algorithm when the torque commands are not 0. Furthermore, when motor torques are positive and varying smoothly, the estimation is highly accurate and it can be used to tell which motor is fault.

B. Vehicle Extreme Cornering Situation

In this test, the vehicle runs on a high adhesion coefficient road and corners continuously. The "Fault Generation" part sets the motor torque T_2 on front right wheel to be 0.25 times of the torque command T_{ci} to simulate the motor failure.

During the test, the vehicle body inertia sensor cluster outputs are shown in Fig. 9. The maximum of vehicle lateral acceleration is nearly ± 10 m/s^2 and maximum of yaw rate is about ± 0.5 rad/s, which means the vehicle is in extreme cornering condition.

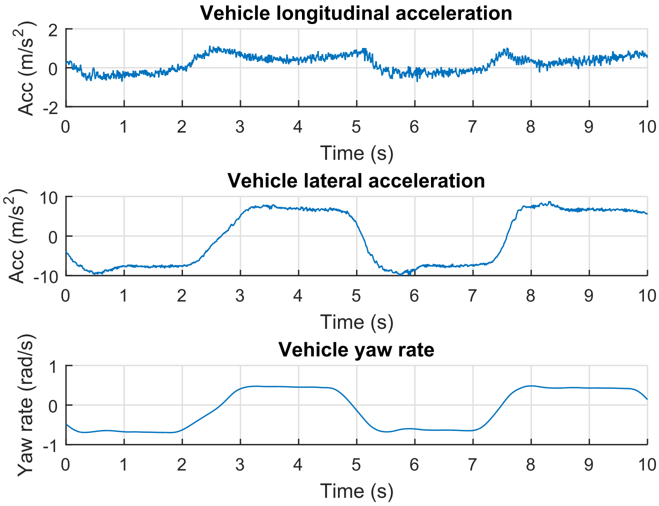

Fig. 9. Vehicle inertia sensor cluster outputs in vehicle cornering situation.

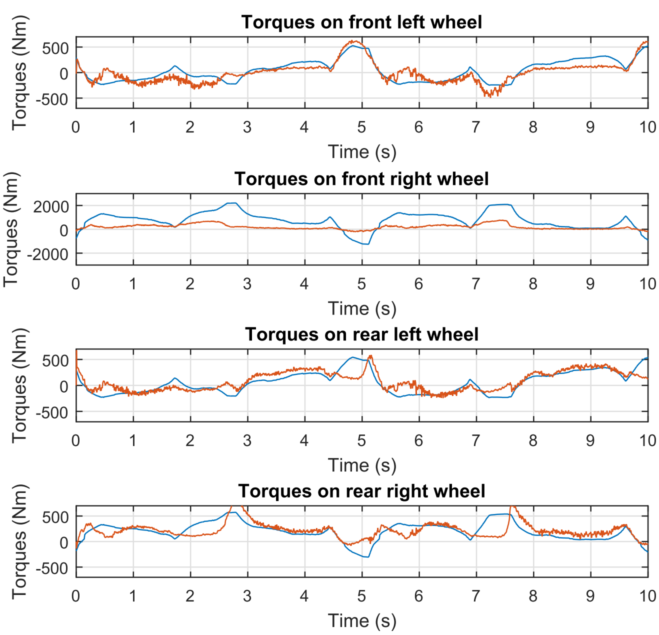

Fig. 10. Motor torque commands and road/tire torques in vehicle cornering situation. (The blue lines are commanded motor torques and red lines are the products of tire/road forces and wheel radii).

With the same method to obtain Fig. 7, motor torque commands and longitudinal tire/road torques are compared in Fig. 10. As the vehicle is in extreme cornering situation and there is vehicle yaw stability control algorithm in VCU to make is stable, the motor torques of the inner wheels and outer wheels are in opposite sign. Besides, the torques are changing fast to maintain the vehicle to follow driver's cornering command. For example, the torque of the front left motor decreases from 500 Nm to −250 Nm in 0.5 s after the time of 5 s. The road/tire torque of the front right wheel is far smaller than the torque command.

The estimated fault parameters $\hat{f}_i$ in the test is shown in Fig. 11. They are oscillating from −1 to 3 during vehicle

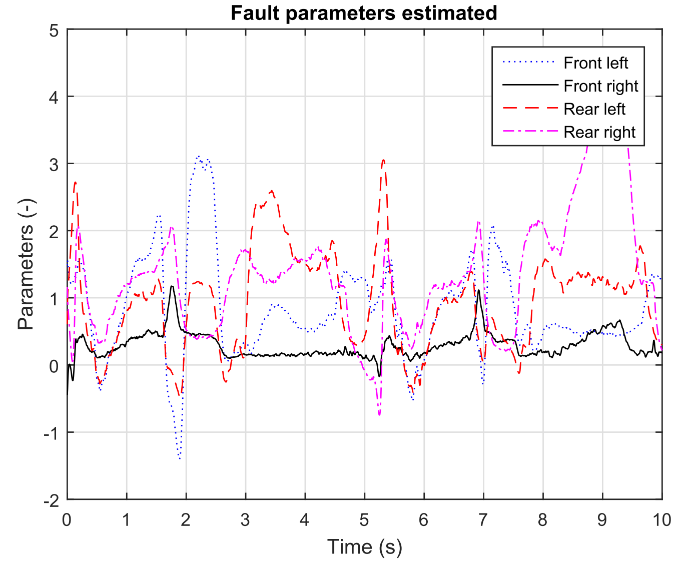

Fig. 11. Fault parameters estimated in vehicle cornering situation.

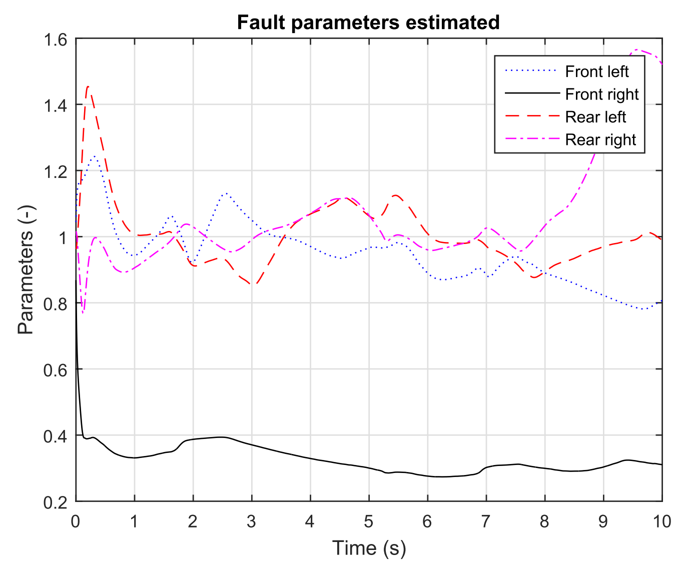

Fig. 12. Fault parameters filtered in vehicle cornering situation.

cornering in extreme situation. To make it clear, a simple Kalman filter is design to make the fault parameters smooth, and the results is shown in Fig. 12. After filtering, the fault parameter $\hat{f}_2$ is around 0.25, which is approximate to the value set by "Fault Generation". The other three fault parameters are around 1, which can be judged as normal.

Based on above two tests, the fault parameters can be estimated in high accuracy when the motor torques change smooth, and they are needed to be filtered when the motor torques change sharply. In the normal vehicle driving situation, such as urban and highway, the motor torques do not change sharply to realize comfortability and efficiency, which implies the diagnosis algorithm is effective in most of vehicle driving situations. In other situations, filter is needed to make fault parameters smooth. Usually, the time is very short and it has small impact on motor diagnosis.

VI. Conclusion

An UKF based motor torque diagnosis algorithm for 4WIMD vehicle is proposed, and the performance of the diagnosis algorithm is evaluated on vehicle tests. The significance of this algorithm is that it is designed based on a nonlinear vehicle model with nonlinear vehicle body dynamics, wheel dynamics and nonlinear tire characteristics, which is valid both in normal and critical vehicle situations. The fault parameters are introduced as unknown parameters to describe the motor faults. As the UKF is suitable for nonlinear estimation problem and there is no need to calculate Jacobin and Hessian matrices, the UKF is implemented to estimate these fault parameters for motor diagnosis purpose.

Two tests with an 4WIMD electric vehicle are carried out to evaluate the performance of the proposed diagnosis algorithm. When the motor torques are stable or change slowly, these fault parameters are estimated accurately. When the motor torques change sharply, the fault parameter varies and a filter is needed to smooth it, and then the fault parameter is accurate. These test results show that the diagnosis algorithm is effective in most of the vehicle driving situations and it could provide motor fault information to fault tolerant controller.

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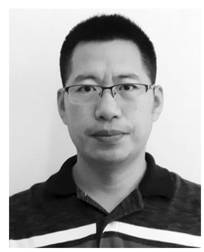

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